QUESTION NO. 22

BANK BRANCH DEPOSIT REPORT

1. Aim

To develop a COBOL program that reads bank transaction records, groups deposits by branch code, displays each branch's total deposit amount, and calculates the grand total of all deposits.

2. Algorithm

1. Start the program.
2. Read the total number of transaction records.
3. Read each fixed-format input record and extract the branch code and deposit amount.
4. For the first record, set its branch code as the current branch.
5. If the branch code changes, display the previous branch total and reset the branch total.
6. Add the deposit amount to the current branch total and the grand total.
7. Repeat until all records are processed.
8. Display the final branch total and the grand total.
9. Stop the program.

3. COBOL Program

```
IDENTIFICATION DIVISION.
PROGRAM-ID. BANK-REPORT.

DATA DIVISION.
WORKING-STORAGE SECTION.

01 INPUT-LINE       PIC X(30).
01 BRANCH-CODE      PIC 9(3).
01 AMOUNT-TEXT      PIC X(9).
01 DEPOSIT-AMOUNT   PIC 9(9).
01 TOTAL-RECORDS    PIC 99 VALUE ZERO.

01 CURRENT-BRANCH   PIC 9(3).
01 BRANCH-TOTAL     PIC 9(12) VALUE ZERO.
01 GRAND-TOTAL      PIC 9(12) VALUE ZERO.

01 DISPLAY-TOTAL    PIC Z(11)9.
01 DISPLAY-GRAND    PIC Z(11)9.

01 RECORD-COUNT     PIC 99 VALUE ZERO.

PROCEDURE DIVISION.

MAIN-PROCESS.

    DISPLAY "Enter number of records:"
    ACCEPT TOTAL-RECORDS

    PERFORM UNTIL RECORD-COUNT = TOTAL-RECORDS
        ACCEPT INPUT-LINE

        MOVE INPUT-LINE(1:3) TO BRANCH-CODE
        MOVE INPUT-LINE(5:9) TO AMOUNT-TEXT

        COMPUTE DEPOSIT-AMOUNT =
            FUNCTION NUMVAL(AMOUNT-TEXT)

        IF RECORD-COUNT = 0
            MOVE BRANCH-CODE TO CURRENT-BRANCH
        END-IF

        IF BRANCH-CODE NOT = CURRENT-BRANCH
            MOVE BRANCH-TOTAL TO DISPLAY-TOTAL

            DISPLAY 'Branch '
                CURRENT-BRANCH ': '
                FUNCTION TRIM(DISPLAY-TOTAL)

            MOVE BRANCH-CODE TO CURRENT-BRANCH
            MOVE ZERO TO BRANCH-TOTAL
        END-IF

        ADD DEPOSIT-AMOUNT TO BRANCH-TOTAL
        ADD DEPOSIT-AMOUNT TO GRAND-TOTAL
        ADD 1 TO RECORD-COUNT
    END-PERFORM

    MOVE BRANCH-TOTAL TO DISPLAY-TOTAL

    DISPLAY 'Branch '
        CURRENT-BRANCH ': '
        FUNCTION TRIM(DISPLAY-TOTAL)

    MOVE GRAND-TOTAL TO DISPLAY-GRAND

    DISPLAY 'Grand total: '
        FUNCTION TRIM(DISPLAY-GRAND)

    STOP RUN.
```

4. Workflow

START
Read the number of transaction records
Read a record and extract branch code and amount
Has the branch code changed?
YES: Display previous branch total and reset it
Add deposit to branch total and grand total
Repeat until all records are processed
Display final branch total and grand total
STOP

5. Sample Input

Enter the record count, then one record per line. The record layout used by the program is: branch code in characters 1–3, one separator character in position 4, and the 9-character amount in positions 5–13.

```
3
101 000001200
101 000000800
205 000002500
```

6. Expected Output

```
Enter number of records:
Branch 101: 2000
Branch 205: 2500
Grand total: 4500
```

7. Result

The COBOL program groups consecutive transaction records by branch, displays the total deposit for each branch, and calculates the grand total of all deposits.

Note: This is a control-break report. Records for the same branch should be consecutive (normally sorted by branch code). The input record positions must match the layout expected by the program.